

PROJECT PRESENTATION • EDITION 2026

# Kohler Architectural Spatial Studio

A generative interior design platform translating room envelopes and financial parameters into code-compliant, AI-curated luxury bathroom suites.

STACK: REACT • THREE.JS • FLASK • GEMINI 2.5 FLASH

# Generative Curation & AI Copilot

Integrating deep learning with architectural spatial logic.

## **Gemini 2.5 Flash Engine**

Synthesizes comprehensive multi-tier product suites (Essential, Curated, Signature) optimized for spatial harmony and aesthetic consistency.

## **In-Studio Copilot Refinement**

Enables real-time natural language directives (e.g., "Shift to Moderne Brass & zero-threshold shower") to dynamically re-curate fixtures.

## **Spatial Constraint Filtering**

Pre-filters raw catalog items against room dimensions (width/depth in feet) and budget caps before LLM generation.

# 3D Visualization & Immersive UI

High-performance spatial rendering and editorial interface design.

## **React Three Fiber Viewport**

Renders real-time 3D models of room layouts and fixture placements directly inside the browser environment.

## **Editorial-Grade Design System**

Styled with bespoke custom cursors, subtle grain overlays, responsive category archives, and high-contrast typography.

## **Optimized Performance Architecture**

Conditional component mounting ensures heavy 3D canvases initialize only after the intro splash sequence finishes.

# Spatial Metrics & Documentation

Rigorous architectural validation and client-ready deliverables.

## **NKBA Clearance Validation**

Automated metrics checking verify mandatory safety tolerances, such as 21-inch egress clearances to adjacent walls.

## **Detailed Fixture Stacks**

Inspectable 4-asset breakdowns covering faucets, toilets, showers, vanities, and bathtubs with deep technical specifications.

## **One-Click PDF Spec Sheets**

Generates professional architectural Bill of Materials (BOM) summary packages ready for client review.

# Production Architecture & Deployment

A robust, decoupled ecosystem built for scalability and performance.

## **Decoupled Client-Server Framework**

Clean separation of concerns between a responsive Vite/React frontend and a robust Python/Flask REST API backend.

## **Local Asset Synchronization**

Predictable static asset pipelines map 50+ high-resolution Kohler product records seamlessly from structured JSON.

## **Version Control & Isolation**

Optimized repository structure with strict .gitignore configurations isolating virtual environments and build outputs.